## IV. Match the following.

1. 

| | |
|---|---|
| Length | kelvin |
| Mass | metre |
| Time | kilogram |
| Temperature | second |

2. 

| | |
|---|---|
| Screw gauge | Vegetables |
| Vernier caliper | Coins |
| Beam balance | Gold ornaments |
| Digital balance | Cricket ball |

## V. Assertion and reason type questions.

Mark the correct answer as:

a. Both A and R are true but R is not the correct reason.

b. Both A and R are true and R is the correct reason.

c. A is true but R is false.

d. A is false but R is true.

1. Assertion(A): The scientifically correct expression is " The mass of the bag is 10 kg"

   Reason (R): In everyday life, we use the term weight instead of mass.

2. Assertion (A): 0 °C = 273.16 K. For our convenience we take it as 273 K after rounding off the decimal.

   Reason (R): To convert a temperature on the Celsius scale we have to add 273 to the given temperature.

3. Assertion (A): Distance between two celestial bodies is measured in terms of light year.

   Reason (R): The distance travelled by the light in one year is one light year.

## VI. Answer very briefly.

1. Define measurement.
2. Define standard unit.
3. What is the full form of SI system?
4. Define least count of any device.
5. What do you know about pitch of screw gauge?
6. Can you find the diameter of a thin wire of length 2 m using the ruler from your instrument box?

## VII. Answer briefly.

1. Write the rules that are followed in writing the symbols of units in SI system.
2. Write the need of a standard unit.
3. Differentiate mass and weight.
4. How will you measure the least count of vernier caliper?

## VIII. Answer in detail.

1. Explain a method to find the thickness of a hollow tea cup.
2. How will you find the thickness of a one rupee coin?

## IX. Numerical Problems.

1. Inian and Ezhilan argue about the light year. Inian tells that it is $9.46 \times 10^{15}$ m and Ezhilan argues that it is $9.46 \times 10^{12}$ km. Who is right? Justify your answer.
2. The main scale reading while measuring the thickness of a rubber ball using Vernier caliper is 7 cm and the Vernier scale coincidence is 6. Find the radius of the ball.
3. Find the thickness of a five rupee coin with the screw gauge, if the pitch scale reading is 1 mm and its head scale coincidence is 68.
4. Find the mass of an object weighing 98 N.



IX_SCI_EM_Unit-01_PHY.indd 12 12-02-2020 14:09:46